

- 1 (a) Different types of respiratory substrate can have different energy values and therefore release different quantities of energy when they are respired.

Complete Table 1.1 to show the energy value of each of the three main types of respiratory substrate.

Use **one** tick (✓) to identify which of the two possible energy values is correct for each respiratory substrate.

Table 1.1

type of respiratory substrate	energy value /kJ g ⁻¹	
	approximately 17	approximately 37
carbohydrate		
lipid		
protein		

[1]

- (b) Determining the respiratory quotient (RQ) of an organism can be used to indicate the main type of respiratory substrate that is being metabolised in respiration. This is because the different types of respiratory substrate have different RQ values. Table 1.2 shows typical RQ values for carbohydrate, lipid and protein.

Table 1.2

type of respiratory substrate	RQ value
carbohydrate	1.0
lipid	0.7
protein	0.8

- (i) State the name of the laboratory apparatus that can be used to determine the RQ value of organisms such as blowfly larvae.

..... [1]

- (ii) When determining RQ values using the laboratory apparatus stated in (b)(i), chemicals such as soda lime or potassium hydroxide solution are used.

State the reason for using chemicals such as soda lime or potassium hydroxide solution when measuring RQ values.

..... [1]

- (c) Organic acids such as malic acid can also act as respiratory substrates. When respired aerobically, their RQ values may be different to the RQ values of the main respiratory substrates.

Fig. 1.1 shows the formula that is used to calculate RQ values.

$$\text{RQ} = \frac{\text{number of molecules of carbon dioxide produced}}{\text{number of molecules of oxygen taken in}}$$

Fig. 1.1

When malic acid is respired aerobically, the equation is:



- (i) Calculate how many molecules of oxygen are taken in when one molecule of malic acid is respired aerobically.

number of molecules of oxygen = [1]

- (ii) Calculate the RQ for malic acid.

Give your answer to **two** decimal places.

RQ = [1]

(d) The deer mouse, *Peromyscus maniculatus*, lives in forests in North America.

Fig. 1.2 shows a deer mouse.



Fig. 1.2

The deer mouse is active throughout the year and is much more active during the night than during the day.

At certain times of the year, deer mice spend a number of hours during the day in a physiologically controlled state of inactivity (**not** active), known as torpor. During this time there is a decrease in metabolic rate.

Fig. 1.3 is a graph showing the RQ of a deer mouse from 6:00 to 22:00 on a day that included time in torpor.

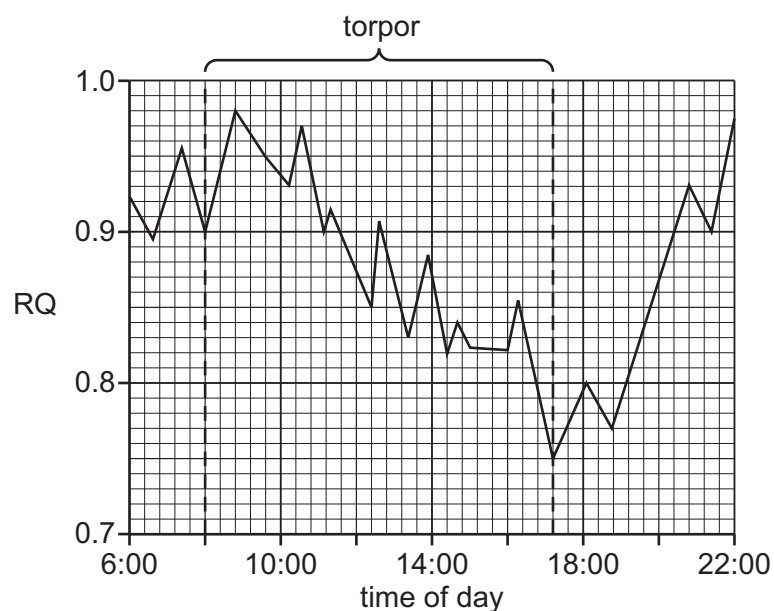


Fig. 1.3



- (i) Describe the trend shown during torpor in Fig. 1.3 **and** suggest an explanation for this trend.

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..... [3]

- (ii) Deer mice have a daily period of torpor only at certain times of the year.

Suggest reasons why a deer mouse enters torpor only at certain times of the year.

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..... [2]

[Total: 10]